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Research Article

Predicting Academic Success and Identifying At-Risk Students: A Systematic Review of Data Analytics and Machine Learning Approaches in Higher Education Institutions

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ABSTRACT

This systematic review aimed to assess the application of data analytics and machine learning techniques in Higher Education Institutions (HEIs) to predict student success and identify those at risk of academic difficulty. While previous research has explored these approaches, a comprehensive synthesis of their effectiveness and limitations has been absent. Moreover, addressing issues such as student attrition and inequitable support is crucial given the global commitment to achieving the Sustainable Development Goals (SDGs), particularly SDG 4, which prioritises inclusive and equitable quality education. A systematic literature review was conducted adhering to Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, encompassing four major databases to bridge this knowledge gap. Studies were included if they utilised data analytics and machine learning techniques to predict student success or identify at-risk students within higher education contexts. A quality assessment using the Critical Appraisal Skills Programme (CASP) Checklist was employed to ensure the rigour of the included studies. The review uncovered a diverse range of approaches, encompassing traditional methods such as logistic regression and support vector machines and more advanced techniques like ensemble methods and neural networks. These approaches were applied to various data sources, including administrative, learning management systems, and survey data. The findings underscored the potential of data analytics and machine learning to revolutionise higher education by facilitating the early identification of at-risk students, tailoring support services to individual needs, and optimising resource allocation. Consequently, leveraging these technologies, HEIs can make data-driven decisions to enhance student success, improve teaching and learning practices, and contribute to attaining the Sustainable Development Goals.

1 Introduction

The global community is confronted with the pressing challenge of achieving the United Nations' Sustainable Development Goals (SDGs) by 2030 (Ncube & Ngulube, 2024). The intricate SDGs encompass multifaceted global issues, including socioeconomic inequalities, environmental degradation and climate instability (Le Blanc, 2014; United Nations, 2015). To eradicate poverty, mitigate climate change and promote quality education, a holistic, interdisciplinary approach is imperative (United Nations, 2015). Higher Education Institutions (HEIs) play a pivotal role in achieving SDG 4, specifically ensuring inclusive, equitable quality education and lifelong learning opportunities (United Nations, 2015). However, HEIs face significant obstacles, notably high student attrition rates, delayed graduation and disparities in academic outcomes among diverse student populations (Aina et al., 2019; Liaison, 2024; Maher & Macallister, 2013; Tamrat, 2022). Traditional methods for identifying students requiring support rely heavily on subjective assessments, limited datasets and retrospective analyses (Ncube & Ngulube, 2024). This can result in delayed interventions, inaccurate identification of at-risk students and inequitable distribution of support services, exacerbating

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existing disparities (Bhutoria, 2022; Hung et al., 2015; Osborne & Lang, 2023). These issues have far-reaching implications for individuals and institutions and for the attainment of SDGs (United Nations, 2015).

Global statistics underscore the pressing concern of student dropout and attrition rates, affecting both developed and developing nations (Tamrat, 2022; UNESCO, 2020). Approximately 20% of students in developing countries abandon higher education programmes. Disparities persist in developed countries, disproportionately impacting specific student demographics (Ressa & Andrews, 2022). These trends emphasize the urgency for innovative strategies to enhance student success and equitable access to higher education.

Data analytics and machine learning offer promising solutions, leveraging systematic examination and analysis of extensive datasets via statistical and computational methods (Abdul-Jabbar & Farhan, 2022; Caspari-Sadeghi, 2023; Sivarajah, 2017; TechTarget, n.d.). Harnessing data-driven insights, Higher Education Institutions (HEIs) can:

- Enhance student progress monitoring,
- Identify at-risk students promptly,
- Allocate resources efficiently (Bozkurt & Sharma, 2022; Nguyen, et al., 2020).

This approach aligns with Sustainable Development Goal 12 (SDG 12), promoting sustainable consumption and production patterns (United Nations, 2015). Data analytics and machine learning optimize resource utilization and minimize waste within HEIs (Ncube & Ngulube, 2024).

The synergistic application of diverse data analytics methodologies, encompassing descriptive, predictive, prescriptive, text and social network analyses, as well as machine learning algorithms and data visualisation, facilitates a multifaceted understanding of student behaviours and academic outcomes (Ncube & Ngulube, 2024). This study focuses on predictive analytics and machine learning, widely utilised for forecasting academic success and identifying at-risk students. Predictive analytics harness historical data to forecast future academic outcomes (Kumar & Garg, 2018; Brown et al., 2015), enabling educators to identify students vulnerable to academic underperformance or dropout (Diamant, 2024; Smithers, 2023). Early detection facilitates targeted interventions, significantly enhancing student success rates (Waheed et al., 2020). Predictive models analyse historical data points, such as academic performance, engagement metrics and course selection, to identify students at risk. Machine learning algorithms analyse extensive student datasets, uncovering patterns and trends indicative of academic achievement or struggle (Jagwani, 2019; Pinto, 2023; Yağcı, 2022). This enables educators to provide personalised support, enhancing student success (Yağcı, 2022).

Numerous data analytics and machine learning approaches have been employed within higher education settings to predict academic success and identify vulnerable students (Järvinen et al., 2021; Rane et al., 2024). Approach selection necessitates careful consideration of research questions, dataset characteristics and desired outcomes. Despite the diversity in these approaches, traditional logistic regression and support vector machines remain prevalent due to their interpretability, efficiency and versatility (Bhumireddy & Anala, 2022; Johnson et al., 2024). Ensemble approaches, combining multiple models, have gained prominence in recent years (Kumari et al., 2018). These methods leverage individual model strengths, mitigating overfitting and enhancing generalisation, thereby improving predictive accuracy (Zhang et al., 2021). Predictive analytics and machine learning within higher education leverage a variety of data sources, including administrative records, educational technology data, gradebook information, and survey responses. Administrative data encompasses student demographics, academic performance and disciplinary records (Kok et al., 2024). Educational technology data from e-books, learning management systems (LMS) and Massive Open Online Courses (MOOCs) provides valuable insights (Asrowi et al., 2019; Reinhold et al., 2020). Institutional data marts integrate multiple sources, analysing student behaviour and predicting academic outcomes (Kurniawan & Halim, 2013). Survey data informs understanding of student perceptions, attitudes and beliefs, enabling prediction of satisfaction and engagement (Chili & Madzimure, 2022; Kandiko-Howson & Matos, 2021).

Ensuring reliability and validity in predictive analytics and machine learning approaches in higher education necessitates careful metric selection (Kroese et al., 2024). While accuracy is commonly employed, its limitations, particularly with imbalanced datasets, underscore the importance of complementary metrics (Hastie et al., 2019; Koehrsen, 2024). A comprehensive evaluation framework incorporates classification metrics such as F1-score, precision, recall and AUC-ROC, which assess model performance, especially with imbalanced datasets (Gonçalves et al., 2014; Luque et al., 2019; Richardson et al., 2024). Additionally, regression metrics like Root Mean Squared Error (RMSE) quantify prediction errors (Sharma et al., 2022). Educational performance metrics, including completion rate, engagement rate and withdrawal rate, provide further insights into model effectiveness and student behaviour (Vărzaru et al., 2021).

While previous research has explored the application of these approaches in HEIs, a systematic review synthesising the existing literature and providing an overview of their application has been lacking. This study

addressed this gap by conducting a review of relevant studies, identifying key findings, and highlighting areas for future research. As such, examining how combined approaches can gain a more comprehensive understanding of student academic challenges and identify at-risk students, this review sought to synthesise existing research on data analytics and machine learning approaches to support students. This synthesis provides valuable recommendations for educators, policymakers, and researchers on optimising data analytics and machine learning approaches for predicting academic success and identifying at-risk students. This study distinguishes itself through transcending theoretical discourse to offer practical recommendations and actionable strategies for implementing data analytics and machine learning approaches in the higher educational context. In this context, the following objectives framed this systematic review:

1 Identify the diverse data analytics and machine learning approaches employed to predict academic success and identify at-risk students in higher education.
2 Examine the various data sources in predictive analytics and machine learning approaches for forecasting academic success and identifying at-risk students in higher education institutions.
3 Explore the implications of data analytics and machine learning approaches for predicting academic success and identifying at-risk students in higher education institutions.

This review concentrated on these objectives as they are vital for informing decision-making among researchers, educators, and policymakers regarding the implementation of data analytics and machine learning. Thus, identifying the most effective approaches enables institutions to select and implement approaches likely to yield optimal results. Relying solely on a single approach can limit effectiveness; thus, understanding the range of available approaches is crucial for institutions. Moreover, recognising the key factors influencing predictive analytics and machine learning approaches' accuracy allows institutions to optimise data collection and analysis processes. Comprehending the implications of data analytics and machine learning in higher education is essential for informed implementation decisions. Consequently, this review's findings contribute to the development of strategies and guidelines governing the use of data analytics and machine learning approaches in HEIs. Through these focused objectives, this review provides a resource for researchers, educators, and policymakers seeking to enhance student outcomes and institutional effectiveness through data-driven approaches.

2 Methods

To maintain methodological rigour and transparency, this systematic review strictly adhered to the reporting guidelines outlined in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) statement (Page, et al., 2021).

2.1 Literature Search and Selection

A systematic search was conducted across four databases, namely, ACM Digital Library, ERIC, Scopus, and Web of Science, to identify relevant literature on predicting academic success and identifying at-risk students in higher education. The search strategy combined keywords and subject headings related to higher education, academic success, at-risk students, data analytics, and machine learning. The following syntax for each database was formulated: ACM Digital Library: (("higher education" OR "universities" OR "colleges") AND (("academic success" OR "academic achievement" OR "student retention") AND ("at-risk students")) AND (("data analytics" OR "learning analytics" OR "educational data mining") OR ("predictive modeling" OR "machine learning algorithms"))). ERIC: ((higher education OR universities OR colleges) AND (academic success OR academic achievement OR student retention OR at-risk students) AND (data analytics OR learning analytics OR educational data mining OR predictive modeling OR machine learning algorithms)) OR TI=("predicting academic success") OR AB=("data analytics" OR "machine learning" OR "higher education"). Scopus: TITLE-ABS-KEY ((((("higher education" OR "universities" OR "colleges") AND ("academic success" OR "academic achievement" OR "student retention" OR "at-risk students")) AND ("data analytics" OR "learning analytics" OR "educational data mining" OR "predictive modeling" OR "machine learning algorithms"))). Web of Science: TS= (("higher education" OR "universities" OR "colleges") AND (("academic success" OR "academic achievement" OR "student retention" OR "at-risk students") AND ("data analytics" OR "learning analytics" OR "educational data mining" OR "predictive modeling" OR "machine learning algorithms. To refine the search, the Population, Intervention, Comparison, Outcome (PICO) framework informed the search strategy and eligibility criteria, yielding peer-reviewed journal articles, conference papers and chapters (Methley et al., 2014). The PICO framework facilitated study selection by guiding the identification of relevant literature. As such, the following eligibility criteria were established:

1.Population (P): Focus was on students enrolled in HIEs, with priority given to graduate students (Master's or Doctoral programmes). While studies on undergraduate students were considered, those exclusively focused on primary, secondary, or vocational education were excluded.

2. Intervention (I): Emphasis was on the application of data analytics and machine learning approaches in higher education. Interventions focused on using data analytics and machine learning approaches for academic success prediction or early warning systems were included. Studies solely focused on the technical development of data analytics without an educational application were excluded.

3. Comparison (C): Attention was on studies comparing the application of data analytics and machine learning interventions to traditional support methods or a control group with no intervention. Baseline measures of academic achievement before intervention are also considered.
4. Outcome (O): Emphasis was on academic success outcomes relevant to higher education students, including academic performance (grades, GPA), student retention and graduation rates, time-to-degree completion, academic progress (credit and course completion), and early warning indicators for at-risk students. Table 1 outlines the additional criteria applied in conjunction with the PICO framework:

Table 1: Additional Eligibility Criteria

Category	Sub-Criteria	Description
Publication	Language	English
	Date	2011 – 2024 (inclusive)
	Type	Peer-reviewed journal articles, conference papers, chapters
Methodology	Type	Quantitative, Mixed methods
	Requirements	Clear and transparent methodology and data analysis
Context	Education Level	Higher education institutions (universities/colleges)
	Data Analytics and Machine Learning Focus	Explicit discussion of data analytics and/or machine learning
Availability	Full-Text	Accessible full-text versions
Intervention	Details	Structure, data types, student support activities
Research	Quality	Demonstrates strong data analytics and machine learning expertise, rigorous methodology, clear sampling, and valuable insights into enhancing student success and identifying at-risk students.

The literature search identified 2414 initial studies. To streamline the review process and ensure high-quality selection, a two-stage approach was implemented using the advanced software tools Rayyan and ASReview (ASReview, 2024; Rayyan, 2024). In the first stage, Rayyan was used to remove duplicate and non-English studies, resulting in 1636 remaining articles (Rayyan, 2024. Subsequently, ASReview analysed these titles and abstracts using pre-defined criteria, prioritising the most relevant studies for full-text review (ASReview, 2024). In the second stage, Rayyan was used to thoroughly review the prioritised articles. Applying eligibility criteria resulted in the exclusion of 1559 studies, identifying 77 key studies that directly addressed the context of the study for further quality assessment.

2.2 Quality Assessment Using the CASP Checklist

A quality assessment was conducted using the Critical Appraisal Skills Programme (CASP) checklist (CASP, 2024) to ensure a rigorous and standardised evaluation of the selected studies. This framework assessed four key dimensions:

1. Conceptual Clarity and Comprehensiveness: Reviewers evaluated the authors' understanding of data analytics and machine learning applications in predicting academic success and identifying at-risk students.
2. Methodological Rigour: The checklist guided the assessment of research design, data collection methods, data analysis techniques, and the mitigation of potential biases in predicting academic outcomes.
3. Sampling Clarity: Reviewers assessed the clarity of the target student population, sampling methods, and justification for sample size and representativeness within higher education settings.
4. Valuable Findings: The checklist focused on the clarity of findings, their alignment with research questions or objectives and methodology, and the depth of discussion regarding implications for academic success and at-risk student identification.

The CASP checklist ensured a systematic evaluation of each article's strengths and weaknesses across these four dimensions. This approach led to the selection of 33 high-quality research articles that employed data analytics and machine learning approaches to predict academic success and identify at-risk students. To ensure a consistent and unbiased screening process, inter-rater reliability (IRR) measures were established. A high concordance rate of 85% and a kappa coefficient of 0.80 demonstrated strong agreement between reviewers. Figure 1 presents the PRISMA flow diagram, illustrating the systematic search and study selection methodology.

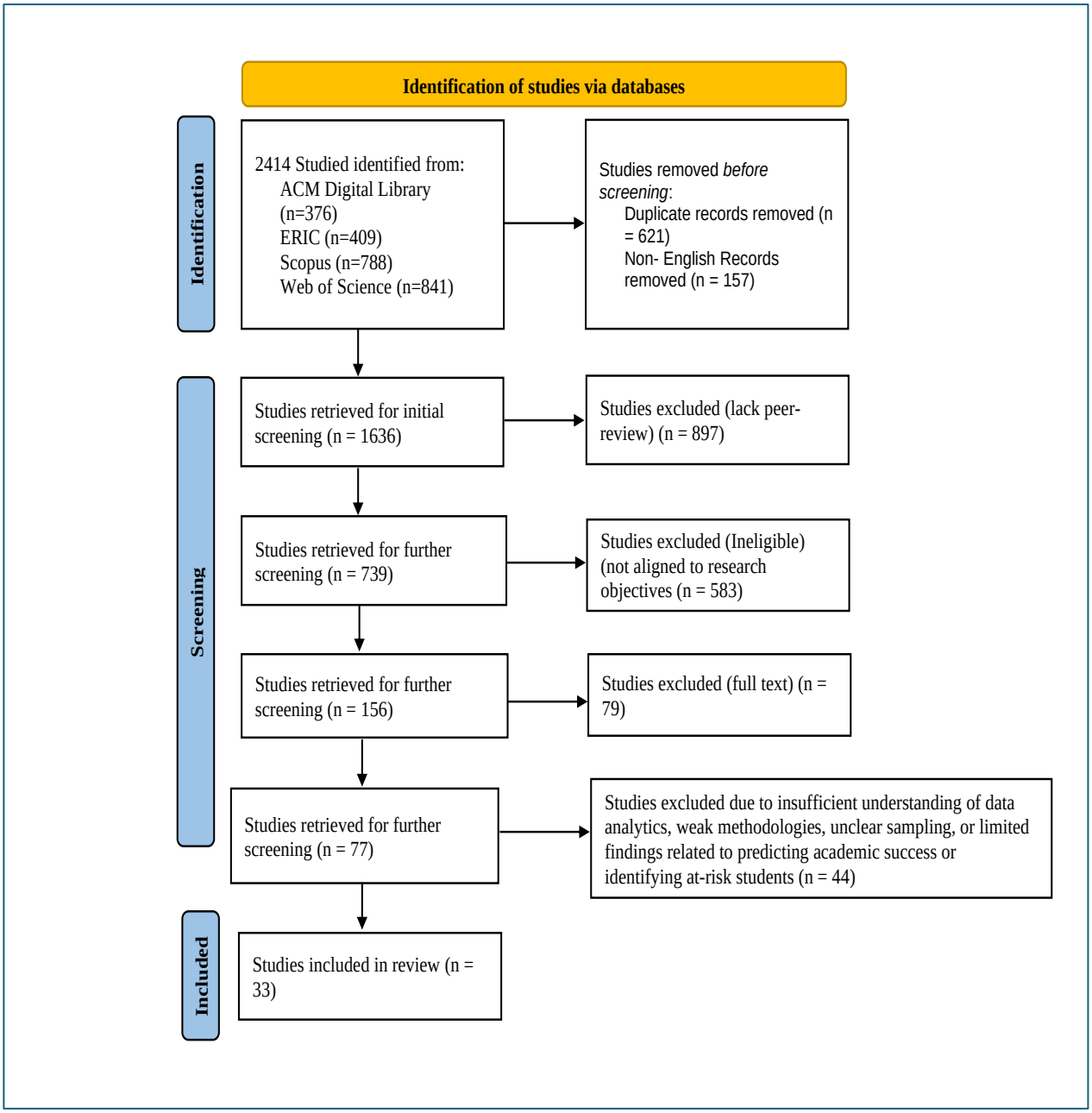


Figure 1: Systematic Literature Search and Selection Process (Adapted from PRISMA 2020 Guidelines)

2.3 Data Extraction and Coding

For methodological rigour, data extraction adhered to PRISMA guidelines (Page et al., 2021). A standardised form developed using CADIMA (2024) ensured consistency and efficiency. The form captured study characteristics, research design, data analytics and machine learning approaches, key findings, and author-identified limitations. Manual coding was then used to extract and document crucial information, including author(s), research design, publisher, and findings. The extracted data is presented in Table 2.

Table 2: Data extraction and Coding

Author(s)	Design	Publisher	Findings
Ahmad et al. (2019)	Quantitative Predictive Analytics Through Case Study and Classification Problem	Springer	Identified significant attributes for student attrition prediction.
Akçapınar et al. (2019)	Quasi-Experimental Design	International Journal of Educational Technology in Higher Education	Accurately predicted unsuccessful students at the end of term (89%). Early prediction (3 weeks) was possible with 74% accuracy.
Akçapınar et al. (2019)	Quantitative Predictive Analytics Design	Smart Learning Environments	Early prediction system using eBook-based teaching-learning.
Alturki, et al. (2022)	Experiment Using Data Mining Algorithms on a Limited Dataset of Easily Accessible Features From Students	Smart Learning Environments	The most significant features are semester grades, distance from accommodation, and culture.
Anagnostopoulos, et al. (2020)	Quantitative Predictive Analytics	Springer	Developed a prediction model using Learning Management System data to identify at-risk students. Improved performance.
Azizah, et al. (2024)	Quantitative Quasi-Experimental	Artificial Intelligence	Analysed motivational variables to identify at-risk students in blended learning. Time-management variables were key factors.
Baashar, et al. (2022)	Quantitative Predictive Analytics	Alexandria Engineering Journal	Predicted academic performance of postgraduate students using ML algorithms. The Artificial Neural Network (ANN) model achieved the best performance (89% R2).
Bainbridge, et al. (2015)	Quantitative Predictive Analytics	Journal of Public Affairs Education	Used learning analytics to identify at-risk students in an online Master of Public Administration programme. Simple models achieved high sensitivity and low false positive rates.
Bañeres, et al. (2020)	Quantitative Predictive Analytics	Applied Sciences	Developed an early warning system to detect at-risk students in online higher education. Used a predictive model, established a threshold, and implemented an early warning system.
Chui, et al. (2018)	Quantitative Predictive Analytics	Computers in Human Behavior	Developed a probabilistic logistic regression model to identify at-risk

			students. Incorporated student engagement data, demographic data, and student performance data.
Herodotou, et al. (2020)	Quantitative Experimental Through Randomised Controlled Trial (RCT)	Journal of Learning Analytics	Used a predictive model to identify at-risk students and implemented motivational interventions. Found that interventions were effective in facilitating course completion.
Hu, et al. (2014)	Quantitative Predictive Analytics Through Longitudinal Study	Computers in Human Behavior	Developed an early warning system using data-mining techniques to predict at-risk students in an online undergraduate course.
Jia, et al. (2015)	Quantitative Predictive Analytics Through Longitudinal Study	Higher Education	Developed a predictive model to identify at-risk students using administrative data.
Koutina, et al. (2011)	Quantitative Predictive Analytics Through an Experimental Study	Springer	Investigated the most efficient machine learning approach for predicting the final grades of postgraduate students.
Latif, et al. (2022)	Quantitative Predictive Analytics Through an Experimental Study	International Journal of Computing and Digital Systems	Predicted student grades using Artificial Intelligence algorithms.
Lourens and Bleazard (2016)	Quantitative Predictive Analytics Through Case Study	South African Journal of Higher Education	Identified second-year dropout students using pre-university information and first-semester data. Used Konstanz Information Miner (KNIME) for predictive modeling.
Matz, et al. (2023)	Quantitative Predictive Analytics Design Through Multi-Institutional Study	Scientific Reports	Predicted student dropout using macro-level (socio-demographics, early performance) and meso-level (engagement data) variables.
Nimy, et al. (2023)	Quantitative Learning Analytics Design Through Multi-Stage Study	Applied Sciences	Developed a probabilistic logistic regression model to identify at-risk students at different stages.
Osborne and Lang (2023)	Quantitative Predictive Analytics Through Binary Classification Problem	Journal of Postsecondary Student Success	Used a neural network model to predict student failure based on grade book data.
Paterson and Guerrero (2019)	Quantitative Predictive Analytics Through Binary Classification Problem	Research in Higher Education Journal	Compared logistic regression and discriminant analysis for predicting student retention.

Pecuchova and Drlik (2023)	Quantitative Predictive Analytics Through Classification Problem	Procedia Computer Science	Evaluated ensemble methods for early prediction of at-risk students.
Pek, et al. (2023)	Quantitative Predictive Analytics Through Classification Problem	IEEE Access	Developed a hybrid ensemble model to predict at-risk students.
Psathas, et al. (2023)	Quasi-Experimental Study	Computers	Examined factors contributing to early prediction of Massive Open Online Courses (MOOCs) dropouts.
Queiroga, et al. (2020)	Quantitative Predictive Analytics Through Case Study and Classification Problem	Applied Sciences	Developed a solution using student interactions with the virtual learning environment to predict at-risk students. Applied a genetic algorithm for hyperparameter tuning.
Qushem, et al. (2024)	Quantitative Predictive Analytics Through Regression/Classification Problem	Technology, Knowledge and Learning	Developed an early warning prediction model using computing science degree performance data.
Ramaswami, et al. (2023)	Quasi-Experimental Study	Journal of Learning Analytics	Evaluated a learning analytics dashboard (SensEnablr) for its effectiveness in impacting student learning.
Russell, et al. (2020)	Quantitative Quasi-Experimental Design	Computers & Education	Examined the effects of a learning analytics platform (Elements of Success) on at-risk students.
Seo, et al. (2024)	Quantitative Predictive Analytics Design, Classification Problem	Heliyon	Developed a prediction model using real-world data from an online university. Identified significant dropout indicators.
Smirani, et al. (2022)	Quantitative Predictive Analytics Through Classification Problem	Scientific Programming	Proposed a Stacked Generalization for Failure Prediction (SGFP) model to improve student results.
Syed Mustapha (2023)	Quantitative Comparative Study Through Regression and Classification Problems	Applied System Innovation	Compared feature engineering and selection methods for predicting student success.
Varade and Gupta (2024)	Quantitative Predictive Analytics Through Classification Problem	Journal of Electrical Systems	Developed a deep neural network model to predict student academic performance.
Wan Yaacob, et al. (2020)	Quantitative Predictive Analytics Through Classification Problem and Experiment	Journal of Physics: Conference Series	Used data mining techniques to predict student dropout in Computer Science.
Yukselturk, et al. (2014)	Quantitative Predictive Analytics Through Classification Problem and Experiment	European Journal of Open, Distance and E-Learning	Predicted dropouts using data mining approaches in an online program.

3 Results

This section synthesises the review's findings through a critical analysis, assessing their relevance and contribution to achieving the research objectives. Specifically, it examines data analytics and machine learning approaches predicting academic success and identifying at-risk students in higher education (Objective 1), summarising results in Table 3.

Table 3: Data Analytics and Machine Learning Approaches for Predicting Academic Success and Identifying At-Risk Students

Author(s)	Approach	Evaluation Metric
Baashar et al. (2022)	Neural Network	Root Mean Square Error
Osborne et al. (2023)	Neural Network	Accuracy
Hu et al. (2014)	Classification and Regression Trees, Adaptive Boosting	Accuracy
Yukselturk et al. (2014)	Data Mining	Accuracy
Varade et al. (2024)	Deep Neural Network	Accuracy
Anagnostopoulos et al. (2020)	Ensemble	F1-score
Matz et al. (2023)	Ensemble	Area Under the Receiver Operating Characteristic Curve
Pecuchova et al. (2023)	Ensemble	Area Under the Receiver Operating Characteristic Curve
Syed Mustapha (2023)	Feature Engineering	Coefficient of Determination
Queiroga et al. (2020)	Genetic Algorithm	Area Under the Receiver Operating Characteristic Curve
Pek et al. (2023)	Hybrid Ensemble	Accuracy
Akçapınar et al. (2019)	k-Nearest Neighbor	Precision, Recall
Herodotou et al. (2020)	Learning Analytics	Completion Rate
Ramaswami et al. (2023)	Learning Analytics	Engagement
Russell et al. (2020)	Learning Analytics	Withdrawal Rate
Seo et al. (2024)	Light Gradient Boosting Machine	Accuracy
Azizah et al. (2024)	Logistic Regression	Coefficient of Determination
Bainbridge et al. (2015)	Logistic Regression	Sensitivity
Jia et al. (2015)	Logistic Regression	Odds Ratio
Paterson et al.	Logistic Regression	Odds Ratio
Qushem et al. (2024)	Logistic Regression	Accuracy
Wan Yaacob et al. (2020)	Logistic Regression	Odds Ratio
Koutina et al. (2011)	Naive Bayes	Accuracy
Lourens and Bleazard (2016)	Predictive Modeling	Sensitivity
Chui et al. (2018)	Probabilistic Logistic Regression	Area Under the Receiver Operating Characteristic Curve
Nimy et al. (2023)	Probabilistic Logistic Regression	Area Under the Receiver Operating Characteristic Curve
Bañeres, et al. (2020)	Quantitative Predictive Analytics	Accuracy
Akçapınar et al. (2019)	Random Forest	Accuracy
Alturki et al. (2022)	Random Forest	Area Under the Receiver Operating Characteristic Curve
Latif et al. (2022)	Sequential Minimal Optimisation, Logistic Regression	Accuracy
Smirani et al. (2022)	Stacked Generalisation	Accuracy
Ahmad et al. (2019)	Support Vector Machines (SVMs)	F1-score
Psathas, et al. (2019)	Support Vector Machines (SVMs)	Recall, Precision

Table 3 presents a comprehensive overview of the diverse data analytics and machine learning techniques that predict student performance. These techniques, from traditional methods like logistic regression to advanced ensemble methods, are applied to various data sources to forecast student outcomes. To evaluate their models, researchers employed a variety of metrics, including classification metrics (F1-score, precision, recall, accuracy, AUC-ROC) and regression metrics (R-squared, RMSE). Additionally, educational performance metrics such as sensitivity, odds ratio, completion rate, engagement, and withdrawal rate were used to assess student outcomes. These metrics provide valuable insights into the accuracy and effectiveness of the models in predicting student performance and identifying at-risk students.

The second research objective of this study was to examine the diverse data sources employed in predictive analytics and machine learning models designed to forecast academic success and identify at-risk students within higher education institutions. To achieve this, the present study analysed the data sources the reviewed studies utilised, summarising the results in Table 4.

Table 4: Diverse Data Sources Employed in Predictive Analytics and Machine Learning Models

Author(s)	Data Source
Ahmad et al. (2019), Alturki et al. (2022), Baashar et al. (2022), Chui et al. (2018), Hu et al. (2014), Jia et al. (2015), Koutina et al. (2011), Latif et al. (2022), Lourens and Bleazard (2016), Nimy et al. (2023), Paterson et al. (2019), Pek et al. (2023), Pecuchova et al. (2023), Qushem et al. (2024), Seo et al. (2024), Smirani et al. (2022), Syed Mustapha (2023), Varade et al. (2024), Wan Yaacob et al. (2020)	Administrative Data
Akçapınar et al. (2019)	eBook-based Teaching
Osborne et al. (2023)	Grade Book Data
Bañeres, et al. (2020)	Institutional data mart containing curated data from six semesters
Anagnostopoulos et al. (2020), Bainbridge et al. (2015), Herodotou et al. (2020), Ramaswami et al. (2023), Russell et al. (2020), Akçapınar et al. (2019)	Learning Management System
Psathas, et al. (2023)	MOOC data, including student demographics, interaction data, and self-reported self-regulated learning (SRL) data
Matz et al. (2023)	Multi-Institutional Data
Yukselturk et al. (2014)	Online Programme Data
Azizah et al. (2024)	Survey Data
Queiroga et al. (2020)	Virtual Learning Environment

Table 4 presents a diverse range of data sources, including administrative data, LMS data, survey data, institutional data marts, and specialised data sources like eBook-based teaching data, MOOC data, and online programme data, which were utilised in the reviewed studies to predict student performance and identify at-risk students. The third objective of this study was to explore the implications of data analytics and machine learning approaches for predicting academic success and identifying at-risk students in higher education institutions. The findings from the reviewed studies regarding this objective are summarised in Table 5.

Table 5: Implications of Data Analytics and Machine Learning for Predicting Academic Success and Identifying At-Risk Students

Key Finding	Implications for Higher Education Institutions	Example Studies
Early Identification of At-Risk Students	Proactive interventions to prevent academic failure and improve student success.	Akçapınar et al. (2019), Alturki et al. (2022), Anagnostopoulos et al. (2020)
Accurate Prediction of Academic Performance	Tailored support services and resource allocation based on individual student needs.	Azizah et al. (2024), Baashar et al. (2022), Bainbridge et al. (2015)
Identification of Diverse Factors Influencing Academic Success	A comprehensive approach to student support that addresses multiple factors contributing to academic performance.	Bañeres et al. (2020), Chui et al. (2018), Herodotou et al. (2020)
Customisation of Models to Institutional Contexts	Adaptability of data analytics solutions to the specific needs of different institutions.	Hu et al. (2014), Jia et al. (2015), Koutina et al. (2011)
Cost-Effectiveness of Data Analytics	Efficient use of resources and improved return on investment.	Latif et al. (2022), Ahmad et al. (2019)

Table 5 underscores the transformative potential of data analytics and machine learning in revolutionising higher education by enhancing student support, improving academic performance, and reducing attrition rates. Firstly, these technologies can identify at-risk students early on, allowing for timely interventions. The table shows that data analytics and machine learning approaches can accurately predict academic performance, enabling tailored support services. Thirdly, they can identify various factors influencing academic success, leading to a holistic approach to student support. Additionally, data analytics models can be customised to suit the specific needs of different institutions. Lastly, these technologies can be cost-effective by optimising resource allocation and improving student outcomes.

4 Discussion

The analysis of study results (Table 3) reveals substantial heterogeneity in the application of data analytics and machine learning approaches across the reviewed studies, highlighting the need to select appropriate approaches based on the specific research problem, dataset characteristics, and desired outcomes in predicting academic success and identifying at-risk students in higher education institutions (Järvinen et al., 2021; Rane et al., 2024). Traditional approaches such as logistic regression and support vector machines have emerged as the most commonly used techniques in the investigated studies. This prevalence can be attributed to their interpretability, efficiency, and versatility in modelling binary outcomes and accommodating various predictor variables (Bhumireddy & Anala, 2022). This finding aligns with existing empirical research (Johnson et al., 2024; Bhumireddy & Anala, 2022), underscoring logistic regression's adaptability, ubiquity, and exceptional suitability for data analytics and machine learning applications focused on predicting academic success and identifying at-risk students. The enduring popularity of these traditional approaches can be attributed to their adaptability, interpretability, and ability to accommodate complex datasets.

The analysis of study results further underscores the widespread adoption of ensemble methods across the examined studies, highlighting their efficacy in predictive analytics for student success, thereby informing targeted interventions and supporting data-driven decision-making within higher education institutions. Consistent with these findings, Kumari et al. (2018) and Zhang et al. (2021) underscore the efficacy of ensemble approaches in augmenting predictive performance in higher education, attributable to their ability to harness the collective strengths of multiple models. This synergistic approach mitigates overfitting and enhances generalisation, optimising student outcome predictions (Kumari et al., 2018). Consequently, ensemble methods have gained widespread acceptance among scholars and higher education institutions, who increasingly recognise their potential to inform data-driven decision-making and improve student success.

Several of the reviewed studies explicitly focus on predictive modelling (Table 3), highlighting its prevalence as a common application of machine learning and data mining approaches, given their ability to identify patterns, extract insights, and make accurate predictions (Cui et al., 2019; Tete et al., 2022). Consistent with the present study's findings, Kuhn and Johnson (2013) and Tete et al. (2022) elucidate that predictive modelling leverages machine learning algorithms to systematically analyse large datasets, thereby uncovering intricate relationships between variables and facilitating evidence-based interventions for at-risk students and data-driven decision-making within higher education institutions. A key finding from the literature review is the effective application of learning analytics in predicting student success and identifying at-risk students by leveraging data about learners and their contextual factors within higher education. Joshi et al. (2020) and Ogata et al. (2022) contend that by harnessing learning analytics, scholars and educators can systematically analyse large datasets to identify early indicators of student struggle, predict academic performance, and inform targeted interventions.

A notable observation from Table 3 is the emergence of neural networks, particularly deep learning architectures, as a prominent approach employed in several studies to address the complex patterns and voluminous datasets inherent in predicting academic success and identifying at-risk students. Characterised by their multi-layered structure, deep learning models have exhibited exceptional proficiency in handling intricate relationships and high-dimensional data (Kovač et al., 2023; Lakkaraju, 2017; LeCun et al., 2015). This proficiency can be attributed to their capacity to learn hierarchical representations, extract salient features, and capture non-linear interactions (Na et al., 2017; LeCun et al., 2015). The application of feature engineering through system innovation in one study is noteworthy, transforming raw data into informative features to enhance predictive models' accuracy in identifying at-risk students and informing targeted support strategies. This approach extracts relevant information from large datasets (Kuhn & Johnson, 2013), reduces dimensionality, and improves model interpretability (Guan et al., 2020). In educational data mining, feature engineering identifies early warning indicators of student struggle (Abdulwahed et al., 2017), predicts student dropout and retention (Ifenthaler & Yau, 2020), and informs personalised learning interventions (Naseer et al., 2024; Rahiman & Kodikal, 2023). Hence, effective feature engineering informs student support by identifying key predictors of academic success, detecting changes in student behaviour, and facilitating timely interventions.

The efficacy of data analytics and machine learning models was contingent upon the judicious selection of evaluation metrics, which quantitatively assessed their performance (Kroese et al., 2024). The reviewed studies (Table 3) reveal a diverse array of evaluation metrics employed across various studies, each with distinct strengths and limitations. Accuracy, the most widely used metric (Table 3), measured the proportion of correctly classified instances. However, as noted by Hastie et al. (2019) and Koehrsen (2024), it may not always provide a comprehensive picture of model performance, particularly in imbalanced datasets. As such, other metrics such as F1-score, precision, and recall were often employed (Table 3). The F1-score, as validated by Powers (2011), offers a balanced measure of precision and recall, providing a more nuanced understanding of model performance. Hastie et al. (2009) further emphasised the complementary nature of precision and recall, with precision measuring the proportion of true positives among all positive predictions and recall measuring the proportion of true positives among all actual positive instances.

For binary classification models, the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) was used to evaluate model performance (Table 3). Gonçalves et al. (2014) and Luque et al. (2019) noted that AUC-ROC plotted true positive rates against false positive rates, providing a thorough evaluation of model performance. Richardson et al. (2024) highlighted the particular usefulness of this metric in evaluating models on imbalanced datasets. Root Mean Squared Error (RMSE) was used for regression models to measure the difference between predicted and actual values, providing an absolute measure of model error (Sharma et al., 2022). In addition to these core metrics, Table 3 showed that other metrics, such as completion rate, engagement, and withdrawal rate, were employed to gain insights into user behaviour and model effectiveness (Vărzaru et al., 2021).

The study revealed diverse data sources employed in predictive analytics and machine learning models within the education sector. As Aldoseri et al. (2023) noted, these data sources are crucial for training and testing models, ultimately leading to more accurate predictions and informed decision-making. Administrative data emerged as the most frequently used data source (Table 4). This aligns with the findings of Kok et al. (2024), who highlighted the importance of administrative data, including student demographics, enrolment records, academic performance data, and disciplinary records, in predicting student success, dropout rates, and academic performance. eBook-based teaching data was another source employed in the reviewed studies (Table 4). This closely relates to the study of Asrowi et al. (2019) and Reinhold et al. (2020), which underscore the widespread use of eBooks in education and the potential of data derived from student interactions with eBooks to predict engagement and learning outcomes. Grade book data, providing insights into student performance on assignments and quizzes, was also utilised by several scholars (Table 4). Latif and Miles (2020) also explicate the extensive use of grade book data to identify at-risk students and provide timely interventions.

Institutional data marts, integrating data from multiple sources, were employed by various studies (Table 4). In that regard, Kurniawan and Halim (2013) elucidate the widespread use of institutional data marts to analyse student behaviour and predict academic outcomes. Learning Management System (LMS) data was also prevalent, providing information about student interactions with online courses (Table 4). Duin and Tham (2020) also underline the extensive use of LMS data to develop predictive student retention and engagement models. Massive Open Online Course (MOOC) data was another data source used to predict academic success and identify at-risk students. The study of He et al. (2015) also concurs with these findings as it notes the wide usage and need of MOOC data, including student demographics, interaction data, and self-reported learning data, to identify factors influencing student success and dropout rates. Online programme data, including information about student participation in online programmes, was also employed by several studies (Table 4). Shou et al. (2024) highlights the role of online programme data in predicting student performance and identifying at-risk students. Survey data, providing insights into student perceptions, attitudes, and beliefs, was another data source utilised (Table 4). Chili and Madzimure (2022) and Kandiko-Howson and Matos (2021) concur with these findings and underscore the need and usage of survey data to predict student satisfaction and engagement.

The study revealed significant implications of data analytics and machine learning for predicting academic success and identifying at-risk students in higher education institutions (Table 5). The studies demonstrated the efficacy of early identification in enabling institutions to proactively implement targeted interventions, such as tutoring and academic advising. This proactive approach contributes to improved student success and retention rates, as supported by Shou et al. (2024) and Chili and Madzimure (2022). The study highlighted the importance of accurate prediction models in tailoring support services to individual student needs. By identifying students most likely to benefit from specific interventions, institutions can optimise resource allocation and ensure that support is provided where it is most needed, as noted by He et al. (2015) and Shou et al. (2024). The study emphasised the need for a holistic approach to student support that addresses multiple factors influencing academic performance. In line with this, Duin and Tham (2020), Kok et al. (2024), and Kurniawan and Halim (2013) highlight the importance of considering demographic information, academic engagement, and personal attributes in developing effective interventions.

Additionally, the study demonstrated the adaptability of data analytics and machine learning solutions to various institutional contexts. In synch with these findings, the studies of Aldoseri et al. (2023) and Duin and Tham (2020) emphasise the importance of customising models to account for specific institutional needs and student populations. The study highlighted the cost-effectiveness of data analytics and machine learning in improving student outcomes (Table 5). Thus, by identifying at-risk students early and providing targeted support, institutions can avoid the costs associated with student attrition and remediation efforts. As such, studies by Latif et al. (2022) and Ahmad et al. (2019) provide evidence of the financial benefits of implementing these technologies.

5 Conclusion

This systematic review delved into the application of data analytics and machine learning approaches in predicting academic success and identifying at-risk students in higher education. The review revealed a diverse array of approaches, including traditional methods like logistic regression and support vector machines and more advanced approaches like ensemble methods and neural networks. These approaches were applied to various data sources, such as administrative, learning management systems, and survey data, to generate valuable insights into student behaviour and performance. The study underscores the potential of data analytics and machine learning to revolutionise higher education. Early identification of at-risk students allows for timely interventions, improving student retention and success rates. Accurate prediction of academic performance enables institutions to tailor support services to individual student needs, optimising resource allocation. A comprehensive approach to student support, informed by data analytics, can address multiple factors influencing academic success, leading to better outcomes. Moreover, data analytics models can be customised to suit the specific needs of different institutions, ensuring their effectiveness in diverse educational contexts. As such, leveraging these technologies, institutions can make data-driven decisions to enhance student success, improve teaching and learning practices, and optimise resource allocation.

6 Limitations and Suggestions for Further Study

This review prioritised the quality of evidence by focusing on peer-reviewed studies. While this approach ensured robustness, it inherently limited the scope of the review. Potentially valuable unpublished research (grey literature) on predicting academic success and identifying at-risk students in higher education might have been excluded. However, the review mitigated this limitation to some extent by employing a comprehensive search strategy across multiple electronic databases to maximise the capture of relevant peer-reviewed studies. Furthermore, the review methods explicitly justified the rationale behind the inclusion/exclusion criteria, particularly the decision to exclude grey literature. The review was inherently limited by the specific inclusion/exclusion criteria used for selecting studies. This may have restricted the generalisability of findings to all education settings. However, by outlining the inclusion/exclusion criteria, the review allowed readers to assess the applicability of the findings to their specific context.

This systematic review provides a solid foundation for understanding the application of data analytics and machine learning in higher education. To further advance this field, future research should explore advanced approaches such as deep learning and natural language processing, address ethical concerns like data privacy and algorithmic bias, and foster interdisciplinary collaboration between computer scientists, educators, and social scientists. Additionally, longitudinal studies can assess the long-term impact of data-driven interventions, and effective data governance frameworks can ensure data quality, integrity, and security. Prioritising these areas, future research can optimise the use of data analytics to enhance student success, improve teaching and learning, and promote equitable and inclusive higher education.

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